Helical Perturbations in Finite-Matter Toroidal–Helical Transport: Nonlinear Distortion, Stability, and Dynamical Recovery

Gerald T. Dahlberg Jr.[1]

[1] *Theoretical and Computational Physics Researcher, Sacramento, California, USA*

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We formulate a nonlinear perturbation framework for a finite-matter toroidal–helical transport background in which the imposed disturbance is characterized explicitly by its amplitude, axial propagation, azimuthal winding, characteristic frequency, and mode number. The aim is to identify parameter ranges in which a helical perturbation undergoes nonlinear wave distortion while remaining bounded, and to determine whether, where, and under what conditions the distorted configuration dynamically recovers toward a reference state. The perturbation is treated as a coupled axial–azimuthal structure rather than as a purely radial or purely axial disturbance, allowing the winding geometry and propagation scale to be varied independently. Stability is assessed through boundedness, excursion magnitude, nonlinear waveform deformation, growth or decay of the perturbation envelope, and recovery after finite displacement. The resulting parameter map is intended to distinguish bounded-recovery, marginal, and unstable regimes and to provide explicit initial-data targets for subsequent nonlinear Einstein–matter evolution. The present analysis does not prescribe the outcome of the full nonlinear evolution; it defines the perturbative variables, diagnostics, and parameter regimes needed to test that evolution systematically.

I. INTRODUCTION

The purpose of this work is to isolate and analyze a helical perturbation imposed on a finite-matter toroidal–helical transport background, with particular emphasis on nonlinear wave distortion, boundedness, and dynamical recovery.

The central question is not whether a prescribed equilibrium exists by construction. Instead, we ask how a finite helical disturbance evolves across a controlled parameter space and whether the perturbed configuration remains bounded or returns toward a reference dynamical state after nonlinear deformation.

The analysis is organized around four explicit control quantities:

$$\mathcal{P} = \{A,\ k_z,\ m,\ \omega\}\,. \tag{1}$$

where A is the perturbation amplitude, k_z controls axial propagation, m is the azimuthal mode number, ω is a characteristic temporal frequency.

The objectives of the paper are:

1. to define the perturbation variables without ambiguity;
2. to quantify nonlinear waveform distortion;
3. to establish boundedness and recovery criteria;
4. to map the stable, marginal, and unstable parameter regimes; and
5. to identify parameter sets suitable as initial-data targets for nonlinear Einstein–matter evolution.

II. TOROIDAL–HELICAL BACKGROUND

We denote the unperturbed background configuration collectively by

$$\mathbf{X}_0 = \mathbf{X}_0(\rho, \phi, z, t), \tag{2}$$

where ρ is a local radial coordinate, ϕ is the azimuthal coordinate, and z parameterizes the axial direction.

The finite-matter toroidal–helical transport background is treated here as a prescribed reference configuration about which the perturbation analysis is carried out.

A useful generic decomposition is

$$\mathbf{X} = \mathbf{X}_0 + \delta\mathbf{X}, \tag{3}$$

with $\delta\mathbf{X}$ representing the finite helical disturbance.

A. Reference state

Define a reference-state diagnostic

$$\mathcal{R}_0 = \mathcal{R}[\mathbf{X}_0], \tag{4}$$

against which the magnitude of departure and subsequent recovery can be measured.

III. HELICAL PERTURBATION VARIABLES

We write the perturbation phase as

$$\Phi(z, \phi, t) = k_z z + m\phi - \omega t + \Phi_0, \tag{5}$$

where k_z controls axial phase progression, $m \in \mathbb{Z}$ controls azimuthal winding, ω is the characteristic temporal frequency, and Φ_0 is an initial phase.

A minimal perturbation field may be written as

$$\delta X = A\, F(\rho)\, \mathcal{W}(\Phi), \tag{6}$$

where $F(\rho)$ specifies radial localization and $\mathcal{W}$ is a waveform that need not remain sinusoidal once nonlinear evolution is included.

At the initial surface one may take

$$\mathcal{W}(\Phi)\big|_{t=0} = \cos\Phi, \tag{7}$$

while allowing nonlinear terms to generate harmonic content dynamically.

A. Winding geometry

The relative importance of axial progression and azimuthal winding may be characterized by a dimensionless winding ratio

$$\eta \equiv \frac{m}{k_z L_z}, \tag{8}$$

where L_z is a chosen axial reference scale.

B. Amplitude measure

For a scalar perturbation, an instantaneous amplitude diagnostic may be defined by

$$A_{\mathrm{eff}}(t) = \max_{\mathbf{x}} |\delta X(\mathbf{x}, t)|\,. \tag{9}$$

IV. NONLINEAR WAVE DISTORTION

The nonlinear problem is distinguished from a purely linear mode calculation by allowing the waveform to deform as the disturbance evolves.

A generic harmonic decomposition is

$$\delta X(\mathbf{x}, t) = \sum_{n=1}^{\infty} a_n(t) F_n(\rho) \cos\left[n\Phi + \theta_n(t)\right]. \tag{10}$$

An initially single-mode perturbation has

$$a_{n>1}(0) \approx 0, \tag{11}$$

whereas nonlinear mode generation produces nonzero higher harmonics.

We define a distortion diagnostic

$$D(t) = \frac{\sum_{n\geq 2} |a_n(t)|^2}{\sum_{n\geq 1} |a_n(t)|^2}, \tag{12}$$

so that $D = 0$ corresponds to an undistorted single-mode signal within this decomposition.

Additional measures may include phase drift, steepening, envelope asymmetry, spectral broadening, and mode-to-mode energy transfer.

V. STABILITY AND DYNAMICAL RECOVERY CRITERIA

The perturbation is classified as bounded over the interval $0 \leq t \leq T$ when

$$\sup_{0\leq t\leq T} A_{\mathrm{eff}}(t) < A_{\max}, \tag{13}$$

for a specified admissible bound $A_{\max}$.

Let a state-distance functional be

$$\mathcal{E}(t) = \|\mathbf{X}(t) - \mathbf{X}_0(t)\|\,, \tag{14}$$

with the norm defined explicitly for the final field variables.

A recovery event occurs when, after an excursion,

$$\mathcal{E}(t_{\mathrm{rec}}) \leq \epsilon_{\mathrm{rec}}, \tag{15}$$

for a chosen tolerance ϵ_{rec}.

The recovery time is then

$$t_{\mathrm{rec}} = \inf\left\{t > t_{\mathrm{peak}} : \mathcal{E}(t) \leq \epsilon_{\mathrm{rec}}\right\}, \tag{16}$$

where t_{peak} is the time of maximum excursion.

We therefore distinguish:

$$\text{bounded recovery:} \quad A_{\mathrm{eff}} < A_{\max}, \quad t_{\mathrm{rec}} < T, \tag{17}$$

$$\text{bounded non-recovery:} \quad A_{\mathrm{eff}} < A_{\max}, \quad t_{\mathrm{rec}} \not< T, \tag{18}$$

$$\text{unstable:} \quad A_{\mathrm{eff}} \geq A_{\max} \text{ or growth is unbounded.} \tag{19}$$

VI. PARAMETER-SPACE MAPPING

The primary scan is performed over

$$(A, k_z, m, \omega). \tag{20}$$

For each parameter point, record at minimum:

$$\{A_{\mathrm{peak}}, D_{\mathrm{peak}}, t_{\mathrm{peak}}, t_{\mathrm{rec}}, \mathcal{E}_{\mathrm{final}}\}\,. \tag{21}$$

The purpose of the scan is to identify surfaces separating:

1. bounded-recovery behavior,
2. bounded but non-recovering behavior,
3. marginal behavior, and
4. unstable growth.

TABLE I. Suggested numerical outputs for each parameter point.

Quantity	Symbol
Initial amplitude	A
Axial propagation parameter	k_z
Azimuthal mode number	m
Characteristic frequency	ω
Peak effective amplitude	A_{peak}
Peak distortion	D_{peak}
Recovery time	t_{rec}
Final state distance	$\mathcal{E}_{\mathrm{final}}$

VII. BRIDGE TO NONLINEAR EINSTEIN–MATTER EVOLUTION

The perturbative analysis is intended to identify well-defined initial-data targets for a subsequent nonlinear Einstein–matter calculation.

The full evolution would require a specified matter model and the coupled system

$$G_{\mu\nu} = 8\pi T_{\mu\nu}, \tag{22}$$

together with the corresponding matter-field equations and constraint conditions.

The present paper does not assume that bounded recovery in the reduced perturbative model guarantees bounded behavior in the full gravitational system. Instead, it identifies parameter sets for which the nonlinear Einstein–matter evolution can be tested under controlled initial conditions.

For each candidate initial-data set, the next-stage analysis should monitor at minimum:

1. constraint violation,
2. curvature growth,
3. stress–energy consistency,
4. trapped-surface or horizon diagnostics where applicable,
5. persistence or loss of the helical structure, and
6. recovery or runaway behavior.

VIII. DISCUSSION

The distinction between nonlinear distortion and instability is important. A waveform may develop substantial harmonic content or geometric deformation while remaining bounded and dynamically recoverable. Conversely, a visually modest distortion may precede secular growth or loss of recovery.

The parameter-space approach therefore avoids identifying waveform deformation itself with failure. The relevant question is whether nonlinear distortion remains inside a bounded basin and whether the perturbed configuration returns toward the chosen reference state.

This framework is intentionally modular. Once the explicit toroidal–helical background and matter sector are fixed, the diagnostics introduced here can be evaluated numerically without changing the logical structure of the stability test.

IX. CONCLUSION

We have defined a working framework for nonlinear helical perturbations on a toroidal–helical background using explicit control variables for amplitude, axial propagation, azimuthal winding, characteristic frequency, and mode structure.

The central outputs are not a predetermined claim of stability, but measurable regimes of nonlinear distortion, boundedness, excursion, and dynamical recovery. The resulting parameter map is designed to identify candidate initial-data sets for subsequent nonlinear Einstein–matter evolution.

The next step is to specify the full background fields and nonlinear evolution equations, then compute the recovery and instability boundaries numerically.